NOTRE DAME UNIVERSITY BANGLADESH

Machine Learning Lab Report-02

Course Code: CSE4214

Course Title: Machine Learning Lab

Lab Task Topic: Linear & Multiple Linear Regression

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Submission Date: January 26, 2026

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1 Objective

The objective of this Machine Learning lab is to understand and implement regression techniques using real-world datasets. This lab focuses on applying Simple Linear Regression and Multiple Linear Regression to analyze the relationship between independent variables and a target variable. The experiments aim to develop predictive models, interpret feature influence, and evaluate model performance using appropriate regression metrics.

2 Dataset Description

Three different datasets are used in this lab to demonstrate the application of regression algorithms.

Dhaka Home Prices Dataset

The `dhaka homeprices.csv` dataset is used for implementing Simple Linear Regression. This dataset contains housing-related information from Dhaka city, where a single independent feature is used to predict house prices. The dataset helps illustrate how a dependent variable varies linearly with one explanatory variable.

Insurance Dataset

The `insurance.csv` dataset is used for implementing Multiple Linear Regression. It contains multiple attributes related to insurance policyholders. The dataset is used to analyze how multiple independent variables collectively influence insurance-related outcomes.

Car Data Dataset

The `cardata.csv` dataset is also used for Multiple Linear Regression. It includes attributes such as `speed`, `car_age`, `experience`, and `risk`. These features are used to predict a target variable by learning the combined effect of multiple factors on the outcome.

3 Linear Regression Implementation

3.1 Loading Dataset for Linear Regression

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

[2]:
```
df= pd.read_csv('dhaka homeprices.csv')
#df = pd.read_csv ('Shopping_cse15_16.csv')
```

[3]:
```
df
```

Explanation

In this step, the required Python libraries are imported to support data handling, visualization, and model preparation. The Pandas library is used to load the dataset `dhaka homeprices.csv` into a DataFrame, which provides a structured tabular representation of the data. This dataset is intended for implementing Simple Linear Regression. Loading the dataset allows for initial inspection and verification of the data before applying further preprocessing and model training.

Output

The output displays the complete contents of the `dhaka homeprices.csv` dataset in tabular form, showing all rows and columns stored in the DataFrame.

[3]:

	area	price
0	2600	55000
1	3000	56500
2	3200	61000
3	3600	68000
4	4000	72000
5	5000	71000
6	2500	40000
7	2700	38000
8	1200	17000
9	5000	100000

3.2 Data Visualization of Home Prices

[4]:
```
plt.xlabel('Area in Square Fit')
plt.ylabel('Price in Taka')

plt.scatter(df['area'],df['price'])
plt.scatter(df['area'], df['price'],color='red', marker='+')

plt.title('Homeprices in Dhaka city')
plt.plot()
```

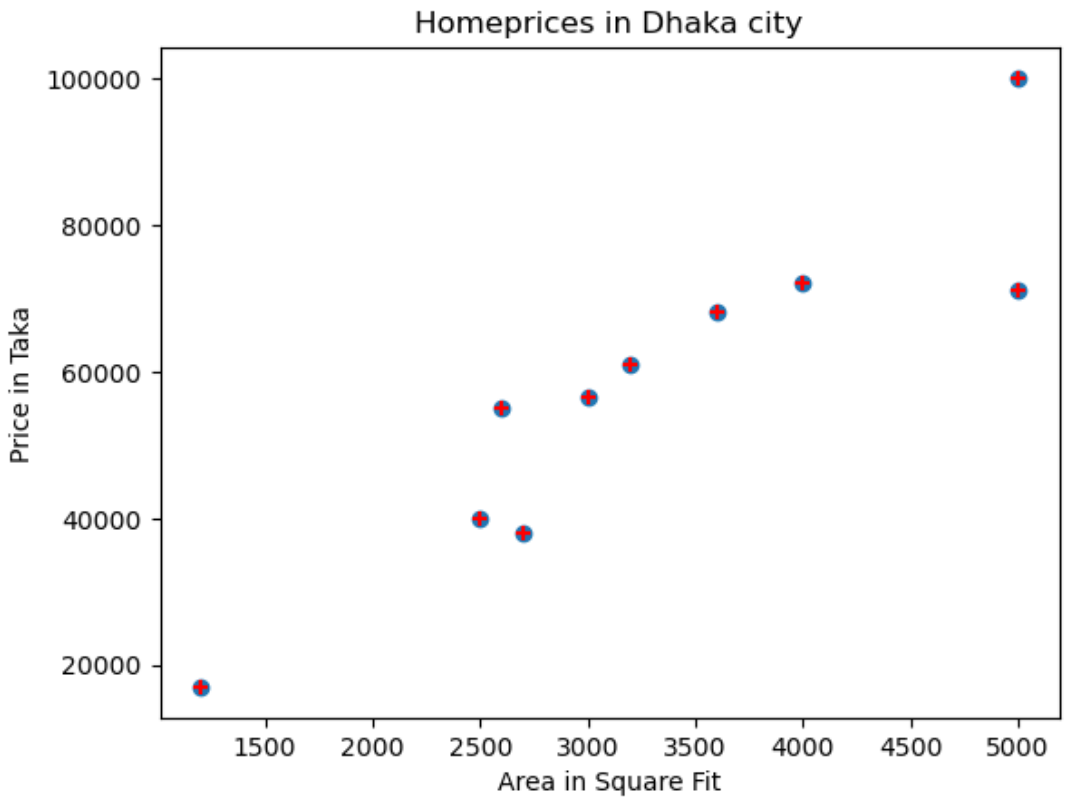
Explanation

This step visualizes the relationship between house area and house price using a scatter plot. The `xlabel()` and `ylabel()` functions are used to label the x-axis and y-axis respectively. The `scatter()` function plots the data points representing house area versus price, where red plus-shaped markers are used for better visibility. The plot title provides contextual information about home prices in Dhaka city. This visualization helps in identifying the linear relationship between the independent and dependent variables.

Output

The output displays a scatter plot showing house prices plotted against area in square feet. The graph indicates an increasing trend, suggesting a positive linear relationship between area and price.

[4]: []



3.3 Splitting Dataset into Training and Testing Sets

[5]:
```
x = df[['area']]
y = df['price']
```

[6]:
```
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size = 0.40,␣
↪random_state =1)

#xtest
#xtrain
```

[7]:
```
xtest



#xtrain
```

Explanation

In this step, the independent variable `area` is assigned to `x`, and the dependent variable `price` is assigned to `y`. The dataset is then divided into training and testing subsets using the `train_test_split()` function. Forty percent of the data is reserved for testing, while the remaining sixty percent is used for training the model. The `random_state` parameter ensures reproducibility of the split. Displaying `xtest` allows verification of the test data samples.

Output

The output displays the feature values of the testing dataset (`xtest`), which contains 40% of the total data selected randomly from the original dataset.

[7]:
```
   area
2  3200
9  5000
6  2500
4  4000
```

[8]:
```
xtrain
```

[8]:
```
   area
0  2600
3  3600
1  3000
7  2700
8  1200
5  5000
```

[9]:
```
ytest
```

[9]:
```
2     61000
9    100000
6     40000
4     72000
Name: price, dtype: int64
```

3.4 Training and Evaluating the Linear Regression Model

[10]:
```
from sklearn.linear_model import LinearRegression
```

[11]:
```
reg= LinearRegression ()
```

[12]:
```
reg.fit(xtrain,ytrain)
```

[12]:
```
LinearRegression()
```

[13]:
```
LinearRegression ()
```

[13]:
```
LinearRegression()
```

[14]:
```
reg.score(xtest,ytest)
```

Explanation

In this step, the Linear Regression model is imported from the `sklearn.linear_model` module and initialized. The `fit()` method is used to train the model using the training data (`xtrain` and `ytrain`). After training, the model performance is evaluated using the `score()` method on the testing dataset (`xtest` and `ytest`). The `score()` function returns the coefficient of determination (R-squared value), which indicates how well the model explains the variance in the target variable.

Output

The output displays the R-squared score of the Linear Regression model. A value closer to 1 indicates better predictive performance, while a lower value indicates weaker model accuracy.

[14]:
```
0.7182056168655753
```

3.5 Prediction Using Trained Linear Regression Model

Explanation

After training the Linear Regression model, the `predict()` method is used to estimate the target variable for new input values. In this step, the model predicts the output for given feature values 3300, 3200, and 2850. These values represent unseen data points, and the trained model applies the learned linear relationship to generate corresponding predictions.

Output

The output consists of three numerical predicted values returned by the regression model. Each value represents the estimated target variable corresponding to the input values 3300, 3200, and 2850, respectively.

[15]:
```
reg.predict([[3300]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
↪X
does not have valid feature names, but LinearRegression was fitted with␣
↪feature
names
  warnings.warn(
```

[15]:
```
array([55021.66064982])
```

[16]:
```
reg.predict([[3200]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
↪X
does not have valid feature names, but LinearRegression was fitted with␣
↪feature
names
  warnings.warn(
```

[16]:
```
array([53572.839244])
```

[17]:
```
reg.predict([[2850]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
↪X
does not have valid feature names, but LinearRegression was fitted with␣
↪feature
names
  warnings.warn(
```

[17]:
```
array([48501.96432364])
```

4 Multiple Linear Regression Implementation-01

4.1 Loading Car Dataset for Multiple Linear Regression

[18]:
```
import pandas as pd
import numpy as np
from sklearn import linear_model
```

[19]:
```
df = pd.read_csv("car data.csv")
```

[20]:
```
df
```

Explanation

In this step, essential Python libraries are imported to support data handling and regression modeling. The Pandas library is used to load and manipulate the dataset, NumPy is utilized for numerical operations, and the `linear_model` module from Scikit-learn provides tools for implementing regression algorithms. The `car data.csv` dataset is then loaded into a Pandas DataFrame using the `read_csv()` function, preparing the data for Multiple Linear Regression analysis.

Output

The output displays the complete contents of the `car data.csv` dataset in tabular form, showing all records and attributes such as `speed`, `car_age`, `experience`, and `risk`.

[20]:

	speed	car_age	experience	risk
0	200	15	5.0	85
1	90	17	13.0	20
2	165	12	4.0	93
3	110	20	NaN	60
4	140	5	3.0	82
5	115	2	8.0	10

[21]:
```
null_values = df.isnull().sum

# Print the result
# print(null_values)
```

[22]:
```
print(null_values)
```

```
<bound method DataFrame.sum of    speed   car_age  experience   risk
0   False    False       False  False
1   False    False       False  False
2   False    False       False  False
3   False    False        True  False
4   False    False       False  False
5   False    False       False  False>
```

[23]:
```
null_count = df.isnull().sum()

# Print the result
print(null_count)
```

```
speed         0
car_age       0
experience    1
risk          0
dtype: int64
```

[24]:
```
df.experience
```

[24]:
```
0     5.0
1    13.0
2     4.0
3     NaN
4     3.0
5     8.0
Name: experience, dtype: float64
```

[25]:
```
df.experience.mean()
```

[25]:
```
np.float64(6.6)
```

[26]:
```
df.experience.median()
```

[26]:
```
5.0
```

[27]:
```
exp_fit= df.experience.median()
```

[28]:
```
exp_fit
```

[28]:
```
5.0
```

[29]:
```
df.experience = df.experience.fillna(exp_fit)
```

[30]:
```
df.experience
```

[30]:
```
0     5.0
1    13.0
2     4.0
3     5.0
4     3.0
5     8.0
Name: experience, dtype: float64
```

[31]:
```
reg = linear_model.LinearRegression()
```

[32]:
```
print(df.columns)
```

```
Index(['speed ', 'car_age', 'experience', 'risk'], dtype='object')
```

[33]:
```
#Strip Leading/Trailing Spaces from Column Names:
#Sometimes column names may have invisible spaces. You can strip any leading/
↪tr
# from the column names using:Strip Leading/Trailing Spaces from Column␣
↪Names:
```

```
df.columns = df.columns.str.strip()
```

[34]:
```
reg.fit(df[['speed', 'car_age', 'experience']], df['risk'])
```

[34]:
```
LinearRegression()
```

[35]:
```
#Predicting risk when speed,car_age and experience are given

reg.predict([[160,10,5]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
↪X
does not have valid feature names, but LinearRegression was fitted with␣
↪feature
names
  warnings.warn(
```

[35]:
```
array([71.37146872])
```

[36]:
```
reg.coef_
```

[36]:
```
array([ 0.33059217,  1.61053246, -6.20772074])
```

[37]:
```
reg.intercept_
```

[37]:
```
np.float64(33.4100009104359)
```

[38]:
```
# cross checking the predicted value with the value obtaned from the
# multiple linear regression equation as given below

160*0.33059217 + 10*1.61053246 + 5*-6.20772074 + 33.410000910435855
```

[38]:
```
71.37146901043586
```

4.1.1 Yes it is the same , Congratulations!!!

5 Multiple Linear Regression Implementation-02

[39]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

[40]:
```
df= pd.read_csv('insurance.csv')
```

[41]:
```
df
```

[41]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200

```
3      33    male  22.705         0     no  northwest  21984.47061
4      32    male  28.880         0     no  northwest   3866.85520
…    …     …     …       …     …        …          …
1333   50    male  30.970         3     no  northwest  10600.54830
1334   18  female  31.920         0     no  northeast   2205.98080
1335   18  female  36.850         0     no  southeast   1629.83350
1336   21  female  25.800         0     no  southwest   2007.94500
1337   61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

5.1 converting categorical values to numeric values

[42]:
```
df['sex'] = df['sex'].astype('category')
df['sex'] = df['sex'].cat.codes
df
```

[42]:
```
      age  sex     bmi  children smoker     region      charges
0      19    0  27.900         0    yes  southwest  16884.92400
1      18    1  33.770         1     no  southeast   1725.55230
2      28    1  33.000         3     no  southeast   4449.46200
3      33    1  22.705         0     no  northwest  21984.47061
4      32    1  28.880         0     no  northwest   3866.85520
…    …  …     …       …     …        …          …
1333   50    1  30.970         3     no  northwest  10600.54830
1334   18    0  31.920         0     no  northeast   2205.98080
1335   18    0  36.850         0     no  southeast   1629.83350
1336   21    0  25.800         0     no  southwest   2007.94500
1337   61    0  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

[43]:
```
df['smoker'] = df['smoker'].astype('category')
df['smoker'] = df['smoker'].cat.codes
df['region'] = df['region'].astype('category')
df['region'] = df['region'].cat.codes
df
```

[43]:
```
      age  sex     bmi  children  smoker  region      charges
0      19    0  27.900         0       1       3  16884.92400
1      18    1  33.770         1       0       2   1725.55230
2      28    1  33.000         3       0       2   4449.46200
3      33    1  22.705         0       0       1  21984.47061
4      32    1  28.880         0       0       1   3866.85520
…    …  …     …       …     …     …       …
1333   50    1  30.970         3       0       1  10600.54830
1334   18    0  31.920         0       0       0   2205.98080
1335   18    0  36.850         0       0       2   1629.83350
1336   21    0  25.800         0       0       3   2007.94500
1337   61    0  29.070         0       1       1  29141.36030
```

```
[1338 rows x 7 columns]
```

[44]:
```
df.isnull().sum()
df
```

[44]:
```
      age  sex     bmi  children  smoker  region      charges
0      19    0  27.900         0       1       3  16884.92400
1      18    1  33.770         1       0       2   1725.55230
2      28    1  33.000         3       0       2   4449.46200
3      33    1  22.705         0       0       1  21984.47061
4      32    1  28.880         0       0       1   3866.85520
…    …  …    …       …       …     …       …            …
1333   50    1  30.970         3       0       1  10600.54830
1334   18    0  31.920         0       0       0   2205.98080
1335   18    0  36.850         0       0       2   1629.83350
1336   21    0  25.800         0       0       3   2007.94500
1337   61    0  29.070         0       1       1  29141.36030

[1338 rows x 7 columns]
```

[45]:
```
x = df.drop(columns='charges')
```

[46]:
```
x
```

[46]:
```
      age  sex     bmi  children  smoker  region
0      19    0  27.900         0       1       3
1      18    1  33.770         1       0       2
2      28    1  33.000         3       0       2
3      33    1  22.705         0       0       1
4      32    1  28.880         0       0       1
…    …  …    …       …       …     …
1333   50    1  30.970         3       0       1
1334   18    0  31.920         0       0       0
1335   18    0  36.850         0       0       2
1336   21    0  25.800         0       0       3
1337   61    0  29.070         0       1       1

[1338 rows x 6 columns]
```

[47]:
```
y = df['charges']
```

[48]:
```
y
```

[48]:
```
0       16884.92400
1        1725.55230
2        4449.46200
3       21984.47061
4        3866.85520
           …
1333    10600.54830
1334     2205.98080
```

```
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1338, dtype: float64
```

[49]:
```
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size = 0.3,␣
↪random_state = 0)
```

[50]:
```
from sklearn.linear_model import LinearRegression
```

[51]:
```
lr= LinearRegression ()
```

[52]:
```
lr.fit(xtrain,ytrain)
```

[52]:
```
LinearRegression()
```

[53]:
```
c = lr.intercept_
```

[54]:
```
c
```

[54]:
```
np.float64(-11827.733141795718)
```

[55]:
```
m = lr.coef_
```

[56]:
```
m
```

[56]:
```
array([  256.5772619 ,   -49.39232379,   329.02381564,   479.08499828,
       23400.28378787,  -276.31576201])
```

5.1.1 Here in the above output , six coefficients for our six training features

[57]:
```
y_pred_train = lr.predict(xtrain)
```

[58]:
```
y_pred_train
```

[58]:
```
array([ 2074.0645306 ,  8141.81393908, 18738.94132528,  7874.86959064,
         6305.12726989,  2023.19725425, 26861.18663021, 14932.93021746,
        10489.56733846, 16254.02800921, 11726.39324257, 11284.0092172 ,
        39312.16870908,  5825.91078917, 12314.92042527,  3164.68427134,
        15406.30681252,  4648.58167988,  5011.79585436,  6012.4796038 ,
        15349.49652486,  8970.97358853,  8780.43012222, 34229.60622887,
         6700.80932636, 26943.25864121, 27280.48004482, 15477.83837581,
         8825.62578924, 34394.38378457, 10177.85528603,  3901.18161227,
        15608.58732963, 29584.76846515, 29453.37088923, 28132.67012427,
        10003.22154888, 33049.08935397,  3963.45204974, 25461.54857001,
         5656.76892592, 27993.86773531,  7049.4472544 , 15100.38851758,
         2552.92266861, 35458.5756605 , 15250.90732084,  3190.28483443,
         1768.85441295, 10155.17603664,  9937.89476088, 11225.91583863,
        16776.25691816,  4332.14442527,  1904.56473771,  4169.01766783,
         5586.26152347,  6181.88067913, 26788.8656339 , 14126.13855797,
        11861.37395532,  7811.00983646, 14043.16898219,  2761.62716836,
```

```
13245.886833  , 11768.08899683,  1979.53264953,  1004.70130715,
36800.01548491,  7337.39948485,  9016.87626313,  2197.43885099,
11522.76560606,  7722.68648352, 11766.49141567, 25673.41065575,
27094.55413975,  8386.55039897,  7851.02612612,   340.5541035 ,
 4812.77154806,  7653.38621355,  3335.8737924 ,  8277.91415433,
 1727.07582017,  4469.3512268 , 33273.43493881,  5960.10203273,
11514.72887612,  9076.67077562, 31445.7194256 , 11381.75279488,
11464.45517614,  4744.19913099,  6959.16143068,  5558.39697169,
 5082.89311652,  9788.2815884 , 31360.93228849,  3182.21328435,
37708.34391043, 10814.52053521,  8936.63917176,  3082.15463971,
11928.53837714, 30612.08362018,  2768.14633037,  5859.36717165,
11271.63384986, 10014.30225729,  6043.73686629, 12975.42245732,
12066.96034099, 28052.15863473, 11361.56357043, 11060.29810116,
14348.98304548, 12312.23589186, 28147.75655535, 11823.51072484,
12848.87384573, 15178.13163272,  4197.13873747, 28308.19263972,
 8971.99085308, 28732.62186124, 11179.06619869,  6579.03176313,
13059.30213716, 14726.65831226,  9783.6398503 ,  2444.05494314,
 5724.67349993,  2786.36526883,  5696.67137344, -1003.82184963,
14605.31776434,  3995.13441381, 10503.83704955, 13075.10394921,
11394.53019512,  9261.88891647, 13608.73784577,  1042.57486857,
11340.66753684,  9273.62568527,  8281.21400296, 14977.26865395,
12554.43628217, 29821.10868267, 17471.58043595, 10459.61280091,
 9389.23502709, 12782.30564709,  5788.18196996, 15693.65157911,
 7291.38867849,  5634.70792056, 31414.11538842, 13661.85253666,
12990.82789567, 12116.96433071, 12483.56884527,  5245.19260841,
 2720.97048488,  3967.45298094,  5831.18485508,   336.78003257,
 8327.6083617 ,  7788.78193696,  5978.53780791, 14780.22108067,
 4154.59712226,  7660.7596099 ,  4986.40913178,   808.63142297,
 6347.28680981,  6028.99252197,  3002.31581222, 30358.06082481,
10242.8271406 , 12160.459422  , 36713.30683683,  5484.52486596,
13900.01813071,  1032.83264035,  7075.58814557, 25622.72563058,
10388.5347597 ,  5748.21820814, 15638.72505346,  8412.71678181,
33607.43410022, 12873.4767783 ,  9089.752017  , 11495.39625664,
29552.08877794,  4833.05771099,  8535.10872516, 10379.54084243,
 2480.53742917, 33644.40927286, 26048.49780558, 12539.77833076,
 7288.62030409,  6523.97395347, 33590.95177474, 34304.29013204,
10903.83077465, 28227.37382203, 33047.75080641,  4484.39001642,
33279.27378584,  9131.498238  , 33409.39957307, 26078.36522976,
13348.90616834, 12332.40769064,  5643.77644808, 15413.40505405,
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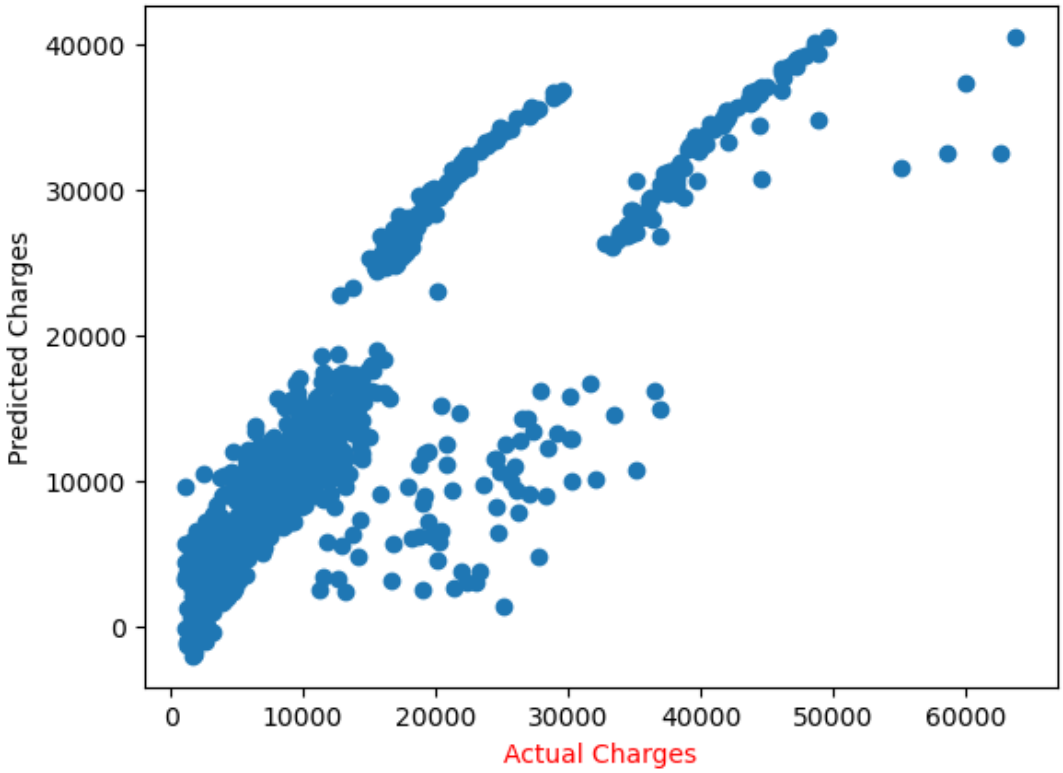
[59]:
```python
import matplotlib.pyplot as plt
plt.scatter(ytrain,y_pred_train)
#plt.xlabel('Actual Charges')
#plt.ylabel('Predicted Charges')
plt.xlabel('Actual Charges', color='red')
plt.ylabel('Predicted Charges', color='black')
plt.show()
```



[60]:
```python
# Plot 'ytrain' using '*'
```

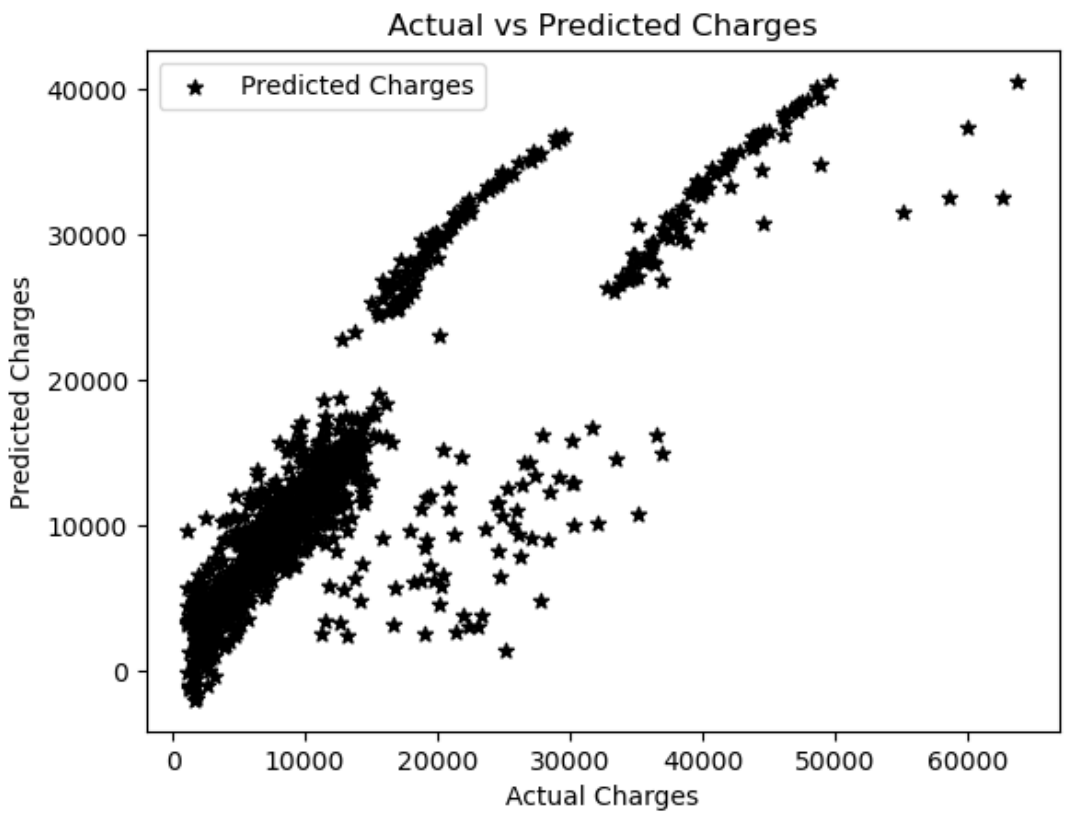
```
# plt.scatter(ytrain, ytrain, marker='*', label='Actual Charges',␣
 ↪color='blue')

# Plot 'y_pred_train' using '+'
plt.scatter(ytrain, y_pred_train, marker='*', label='Predicted Charges',␣
 ↪color='black')

plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.title('Actual vs Predicted Charges')

# Add legend to differentiate between actual and predicted charges
plt.legend()

plt.show()
```



[61]:
```
from sklearn.metrics import r2_score
r2_score (ytrain, y_pred_train)
```

[61]: 0.7306840408360217

[62]:
```
y_pred_test = lr.predict(xtest)
```

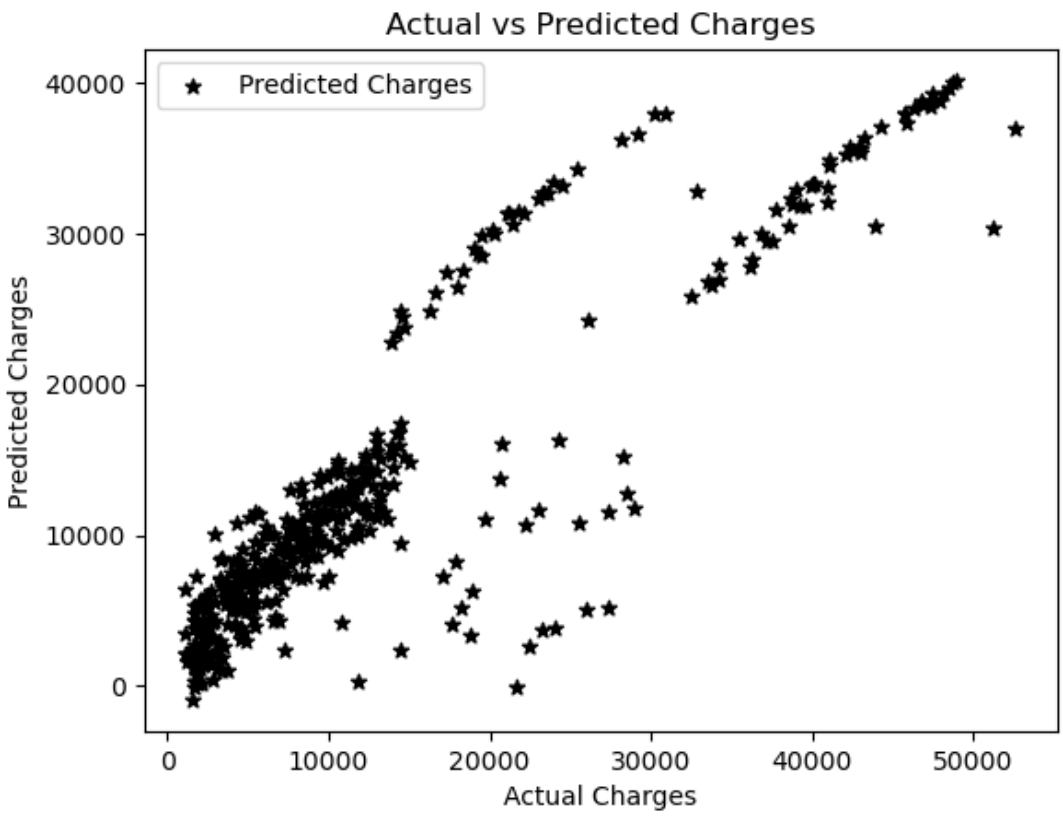
[63]:
```
# Plot 'ytest' using '*'
#plt.scatter(ytest, ytest, marker='*', label='Actual Charges', color='blue')

# Plot 'y_pred_train' using '+'
plt.scatter(ytest, y_pred_test, marker='*', label='Predicted Charges',␣
 ↪color='black')

plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.title('Actual vs Predicted Charges')

# Add legend to differentiate between actual and predicted charges
plt.legend()

plt.show()
```



[64]:
```
r2_score (ytest, y_pred_test)
```

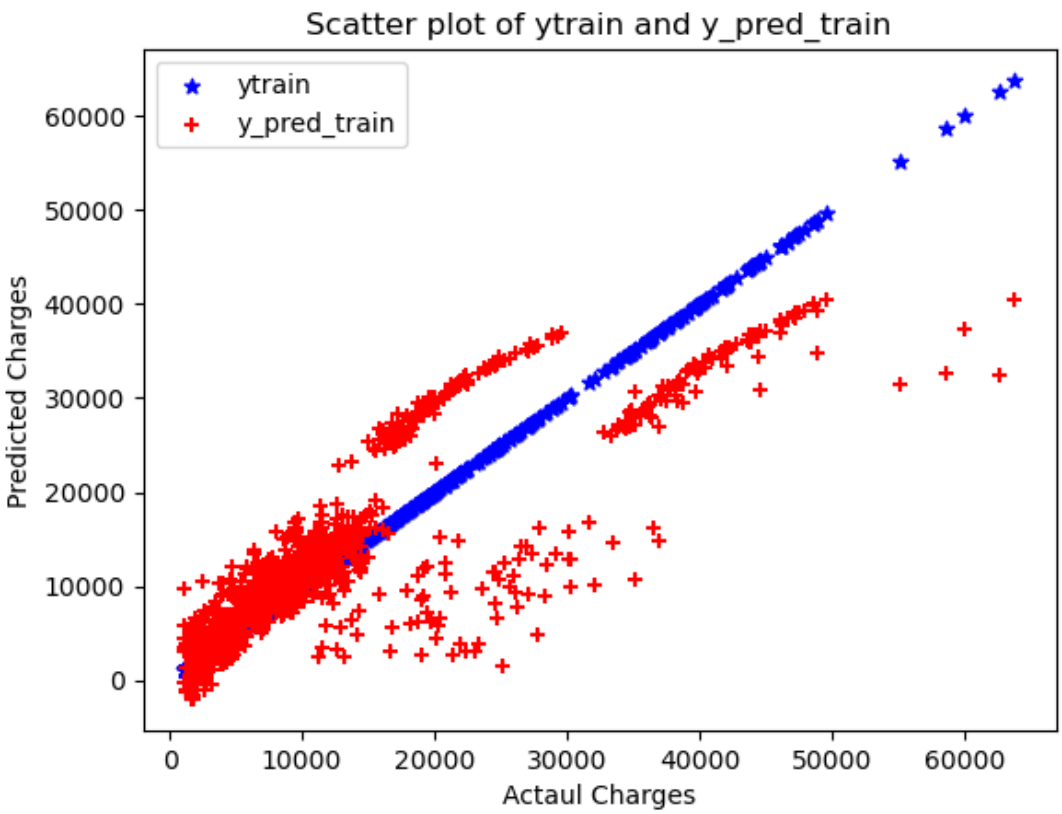
[64]: 0.7911113876316933

[65]:
```
# Create scatter plot
plt.scatter(ytrain, ytrain, marker='*', label='ytrain', color='blue') #␣
 ↪ytrain w
```

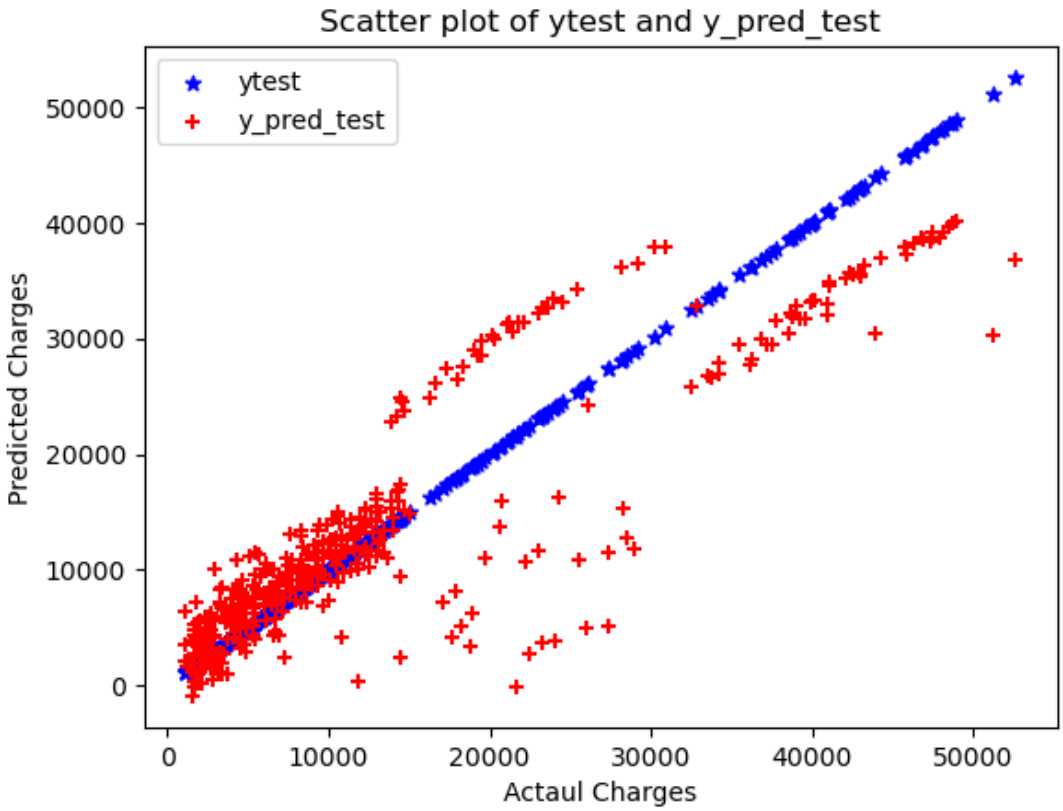
```
plt.scatter(ytrain, y_pred_train, marker='+', label='y_pred_train',␣
 ↪color='red') # y_p

# Add labels and legend
plt.xlabel('Actaul Charges')
plt.ylabel('Predicted Charges')
plt.title('Scatter plot of ytrain and y_pred_train')
plt.legend()

# Show plot
plt.show()
```



[66]:
```
# Create scatter plot
plt.scatter(ytest, ytest, marker='*', label='ytest', color='blue') # ytrain␣
 ↪with
plt.scatter(ytest, y_pred_test, marker='+', label='y_pred_test',␣
 ↪color='red') # y_pred

# Add labels and legend
plt.xlabel('Actaul Charges')
plt.ylabel('Predicted Charges')
plt.title('Scatter plot of ytest and y_pred_test')
plt.legend()
```

```
# Show plot
plt.show()
```